Cardio-respiratory recording

Seismocardiography (chest vibration) and chest-wall breathing motion, 10-minute capture

CardioResp-A1B2C3

Chest motion sensor, 200.0 Hz

Name	Simulated adult (test data)	Patient ID	SIM-001
Age	35 years - Adult (18-39 years)	Sex	not stated
Sensor placement	front (sternum)	Posture	lying on back
Recorded	2026-09-26 09:20:00 - 09:30:00	Duration	10.0 min
Notes	Generated by the PC physiological simulator: 18 s breath-hold, movement and a 40 degree posture change are included.		

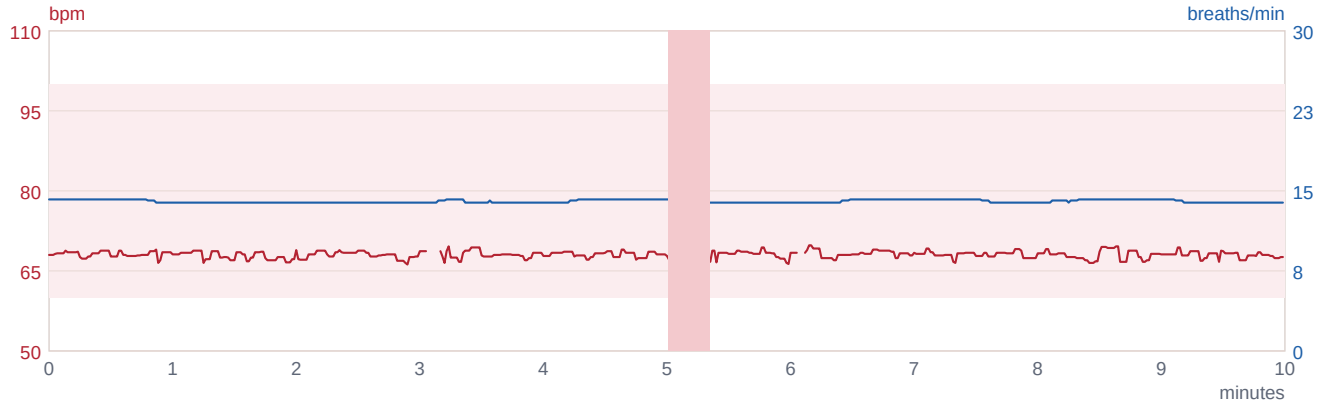
Summary

Measure	Value	Reference for age	Assessment
Heart rate, mean	**68 bpm**	60-100 bpm	within range
Heart rate range (5th-95th pct.)	**67-69 bpm**		
RMSSD / SDNN / pNN50	**31 / 24 ms / 9 %**	RMSSD >= 20 ms	within range
Beats analysed (accepted)	**664 (99 %)**		
Breathing rate	**13.5 /min**	12-20 /min	within range
Breathing cycle variation	**2 %**	< 35 %	within range
Inspiration / expiration	**1.9 s / 2.4 s (1:1.3)**	about 1:1 - 1:3	within range
Breathing depth (chest tilt)	**1.20 deg (variation 1 %)**	relative	
Pauses >= 10 s	**1 (longest 20 s)**	none	**present**
Sighs / posture changes	**0 / 1**		
Resp. sinus arrhythmia	**4.7 bpm per breath**		
Heart beats per breath	**5.0**	about 4-5 (adult)	
Movement excluded	**14 s**		
Signal quality heart / breathing	**96 / 96 of 100**	>= 60 good	

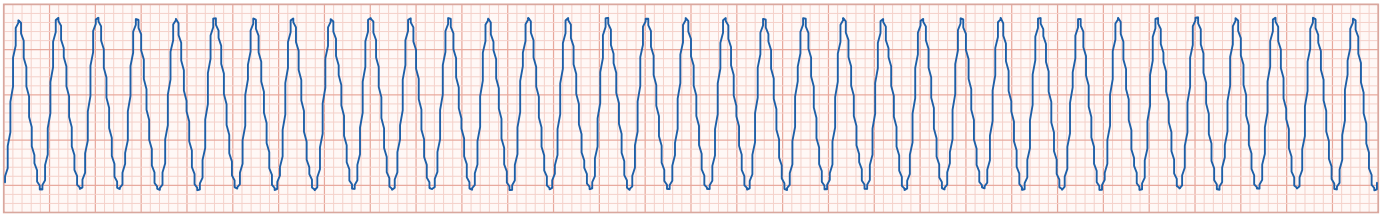
Automated interpretation

- Mean heart rate 68 bpm, within the resting range for adult (18-39 years) (60-100 bpm).
- Short-term HRV: RMSSD 31 ms, SDNN 24 ms.
- Breathing rate 13.5 /min, within the resting range for this age (12-20 /min).
- 1 breathing pause of 10 s or longer (longest 20 s).
- Respiratory sinus arrhythmia 4.7 bpm per breath; 5.0 heart beats per breath.
- Mean signal quality: heart 96/100, breathing 96/100.

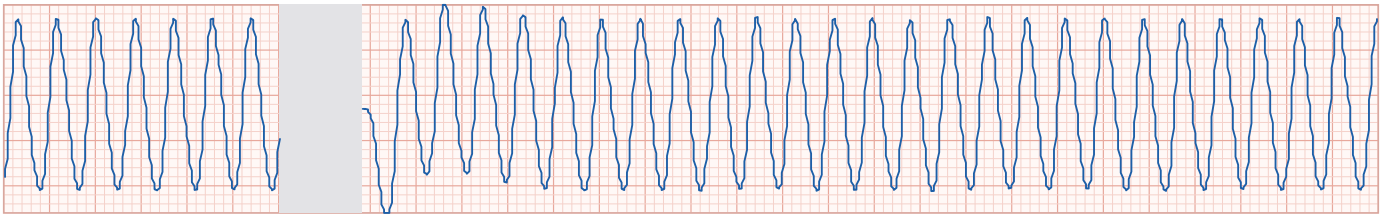
Heart and breathing rate, per second



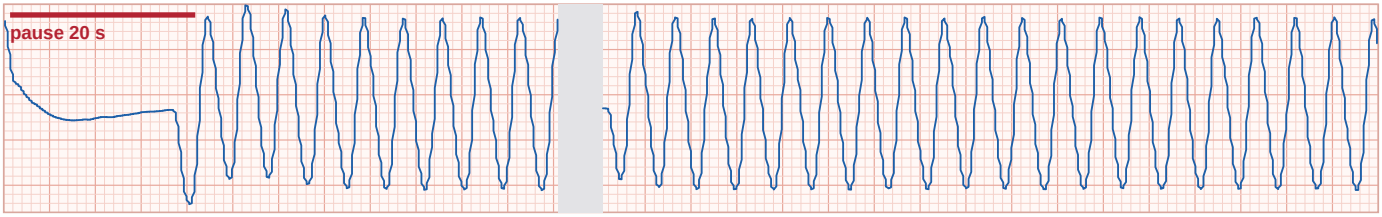
Breathing waveform - chest-wall tilt, last 10 minutes



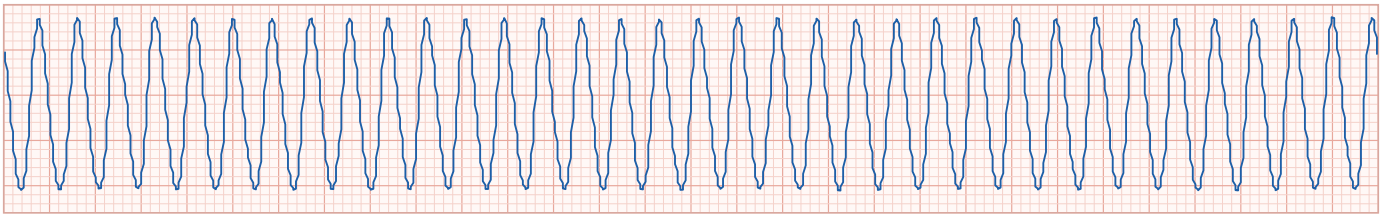
09:20:00 +150 s, full height 1.46 deg, grid 1 s / 5 s



09:22:30 +150 s, full height 1.46 deg, grid 1 s / 5 s



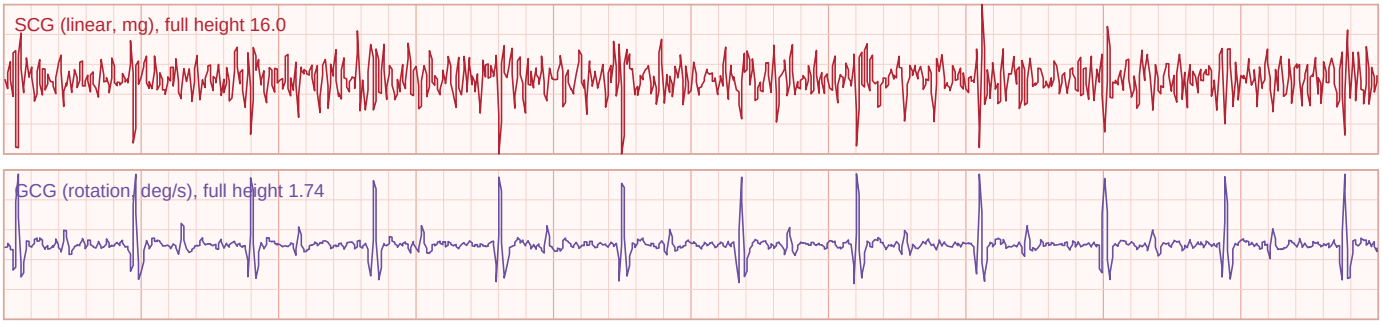
09:25:00 +150 s, full height 1.46 deg, grid 1 s / 5 s



09:27:30 +150 s, full height 1.46 deg, grid 1 s / 5 s

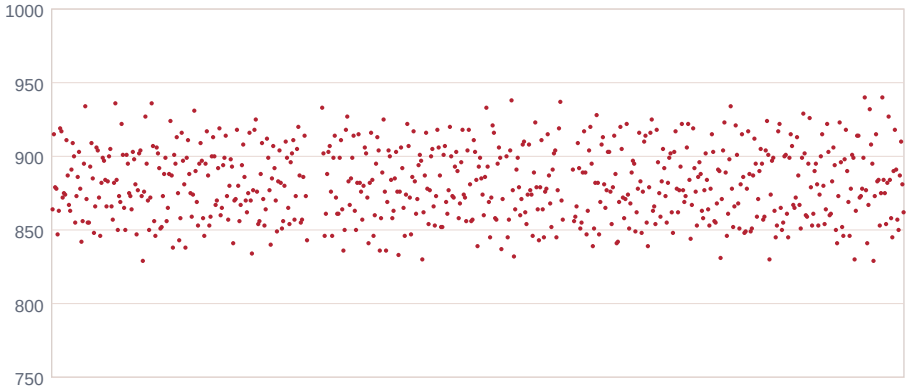
Upward = inspiration. Grey = movement (excluded). Red bars = breathing pauses.

Heart vibration - last 10 seconds (100 Hz)

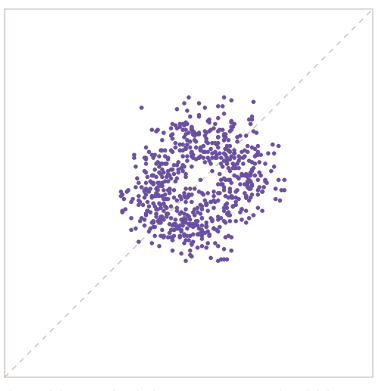


Dominant-axis band-pass (5-30 Hz) seismocardiogram and gyrocardiogram. Grid 0.2 s / 1 s.

Beat-to-beat intervals (ms)



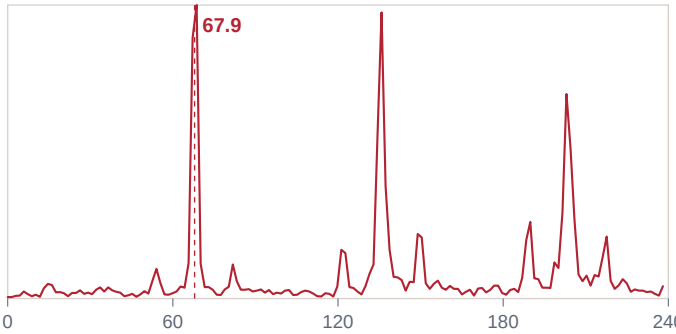
Poincare plot



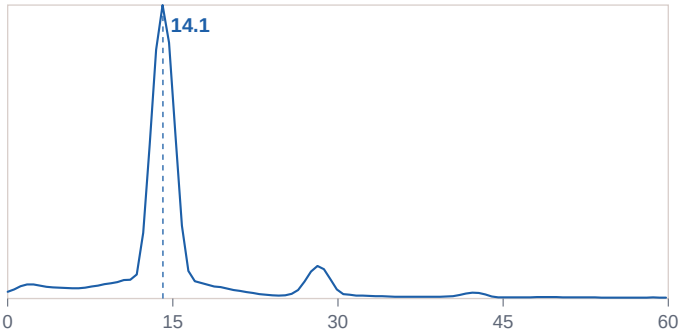
SD1 22 ms, SD2 27 ms, axes 750-1000 ms

Frequency analysis

Heart-beat rhythm (beats/min)



Breathing (breaths/min)



Events in this recording

Time	Event	Duration
09:23:00	Movement	9.1 s
09:25:00	**Breathing pause**	20.2 s
09:26:00	Movement	4.9 s
09:26:06	Posture change	

Method

A chest-worn 6-axis motion sensor (accelerometer and gyroscope) sampled at 200.0 Hz (range +-2 g and +-250 deg/s, 44 Hz anti-alias filter) was fixed to the chest wall. At start-up the gravity direction and gyroscope offsets were measured during 8 s of stillness; the gravity vector is then tracked by fusing gyroscope rotation with the accelerometer.

Breathing is the small rotation of the chest wall relative to gravity. It is band-pass filtered for the patient's age, projected on its principal 3-D axis and segmented breath by breath (inspiration, expiration, depth). The breathing rate is the number of breaths per minute of movement-free time, so pauses lower it.

Heart beats are detected from chest vibration (seismocardiogram, 5-30 Hz) on the accelerometer and from chest rotation (gyrocardiogram) on the gyroscope. Each sensor's dominant axis is found adaptively, the two energy envelopes are fused by periodicity-based quality weights and beats are detected with adaptive thresholds, refractory period, search-back and interval validation. Heart rate is cross-checked by autocorrelation and spectral analysis.

Movement is detected from gravity-free linear acceleration, fast chest rotation and large turns, so breathing itself (also deep or fast breathing) is not taken for movement; affected segments are excluded. When strong breathing shakes the chest, the heart rate is paused instead of locking onto the breathing rate. Coughs are counted only as short explosive vibration bursts with quiet before and after (body movements and taps are rejected), and illness patterns are only suggested after 5-30 minutes of calm monitoring. Reference ranges are awake resting values for the entered age (PALS/AHA); interpretation text is automated and must be reviewed by a clinician.

Reviewed by (clinician)

Date / comments

CardioResp is a research and wellness instrument and not a certified medical device. It does not replace ECG, pulse oximetry, polysomnography or clinical examination. Readings depend on sensor contact, placement and stillness.